

■ Environmental Chemistry

Water Quality Testing · Grade 6 STEM Lesson

1. What Is Environmental Chemistry?

Environmental chemistry is the study of chemicals in nature and how they affect our environment. Scientists examine how substances move through air, water, and soil, and how they impact plants, animals, and people.

Think of it as being a **detective for nature!** Environmental chemists investigate what happens when different chemicals enter rivers, forests, and oceans. They help us understand how to keep our planet healthy and safe for all living things.

2. Why Do We Test Water?

**2
Billion+**

People worldwide lack access to safe drinking water. Testing helps identify harmful contaminants *before* they cause illness. Regular testing ensures the water we drink, cook with, and use daily is safe for everyone.

3. What Is Water Quality?

Water quality tells us how healthy water is based on **three key aspects**:

- | | |
|---------------------|---|
| ■ Physical | How water looks, smells, and tastes — colour, clarity, odour. |
| ■ Chemical | Substances dissolved in water such as minerals, pesticides, heavy metals. |
| ■ Biological | Living organisms like bacteria and viruses present in water. |

4. Common Water Contaminants

■ ■ Chemicals

Pesticides and industrial waste that pollute rivers and groundwater.

■ Microorganisms

Bacteria and viruses that cause serious illnesses like cholera and typhoid.

■ ■ Heavy Metals

Lead and mercury — toxic to human bodies even in tiny amounts.

■ ■ Physical Debris

Plastics and sediments that block water flow and harm aquatic life.

5. How Water Quality Affects Health

■ Clean Water Benefits

- Keeps bodies healthy and strong
- Prevents cholera, typhoid, diarrhoea
- Supports digestion, skin health, immunity

■■ Contaminated Water Risks

- Stomach illnesses, vomiting, diarrhoea
- Skin rashes and problems
- Long-term organ damage from heavy metals

6. Basics of Water Testing

Step 1 — Collect Sample

Gather water from your source using a clean container. Do not touch the inside of the container to avoid contamination.

Step 2 — Use Test Tools

Apply pH strips, turbidity tubes, or chemical test kits to your water sample. Follow instructions carefully.

Step 3 — Record & Analyse

Write down all measurements and observations. Compare results to safe water standards to decide if the water is safe.

Key factors examined: Colour & clarity · Smell · pH level · Chemical content · Microbes

7. Tools for Water Testing

■ pH Strips

Change colour to show if water is acidic or alkaline.

■ Turbidity Tube

Measures how clear the water is.

■ Bacterial Test Kit

Detects harmful bacteria that cannot be seen.

■ Chemical Test Strips

Check nitrate and chlorine levels.

8. Reading Test Results

Parameter	Safe Range	What It Means
pH Level	6.5 – 8.5	Below 7 = acidic · 7 = neutral · Above 7 = alkaline
Turbidity	0 – 1 NTU (clear) < 4 NTU (safe)	Measures water clarity; above 4 NTU may be unsafe
Chlorine	0.2 – 4 mg/L	Disinfects water; too much or too little is harmful

Nitrates	< 10 mg/L	From fertilisers; high levels dangerous for infants
Lead	< 0.01 mg/L	Heavy metal; toxic even in tiny amounts

9. What To Do If Water Is Contaminated?

■ Inform Adults & Authorities

Tell parents, teachers, or local health officials immediately. They can alert the community and coordinate action.

■ Use Filters or Boil Water

Filter water through a clean cloth or purifier. Boiling water for at least 1 minute kills harmful germs.

■ Avoid Unsafe Water

Never drink or cook with water you suspect is contaminated. Join local clean-up efforts to remove pollution from water sources.

10. Benefits of Clean Water

of childhood diseases — including diarrhoea, cholera, and typhoid — can be prevented by clean water.

80%

When families have clean water: children attend school regularly, parents can work without illness, and communities thrive together. **Every drop of clean water matters!**

■ Let's Review!

1. What is one common water contaminant, and why is it dangerous?

Answer: _____

2. Name a tool used to test water pH and explain how it works.

Answer: _____

3. Why is clean water important for community health?

Answer: _____

4. What should you do if you discover contaminated water at your school?

Answer: _____

5. What does a turbidity reading above 4 NTU tell us about the water?

Answer: _____

■ *Keep testing and protecting our water! · Grade 6 STEM · Environmental Chemistry*